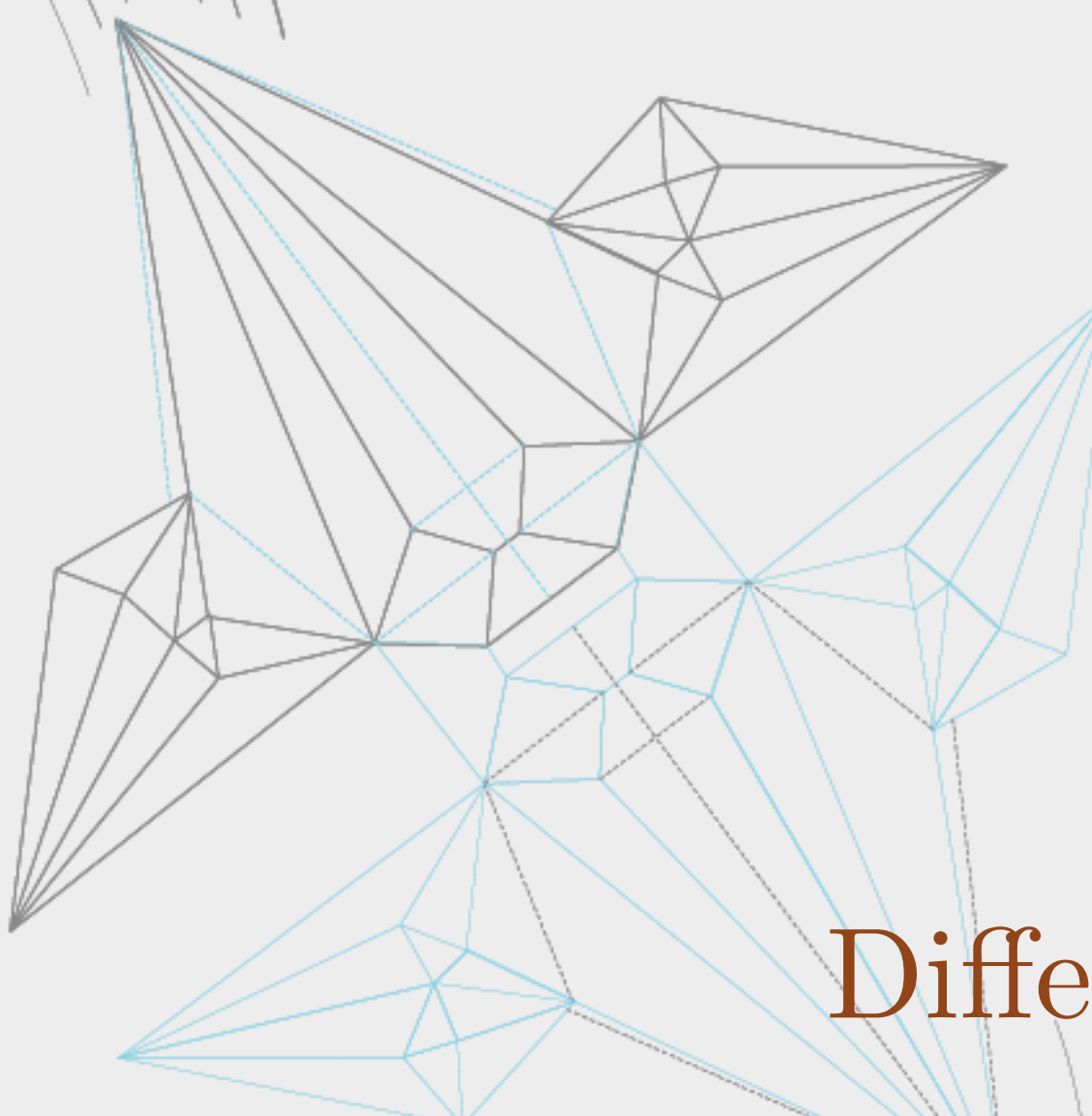




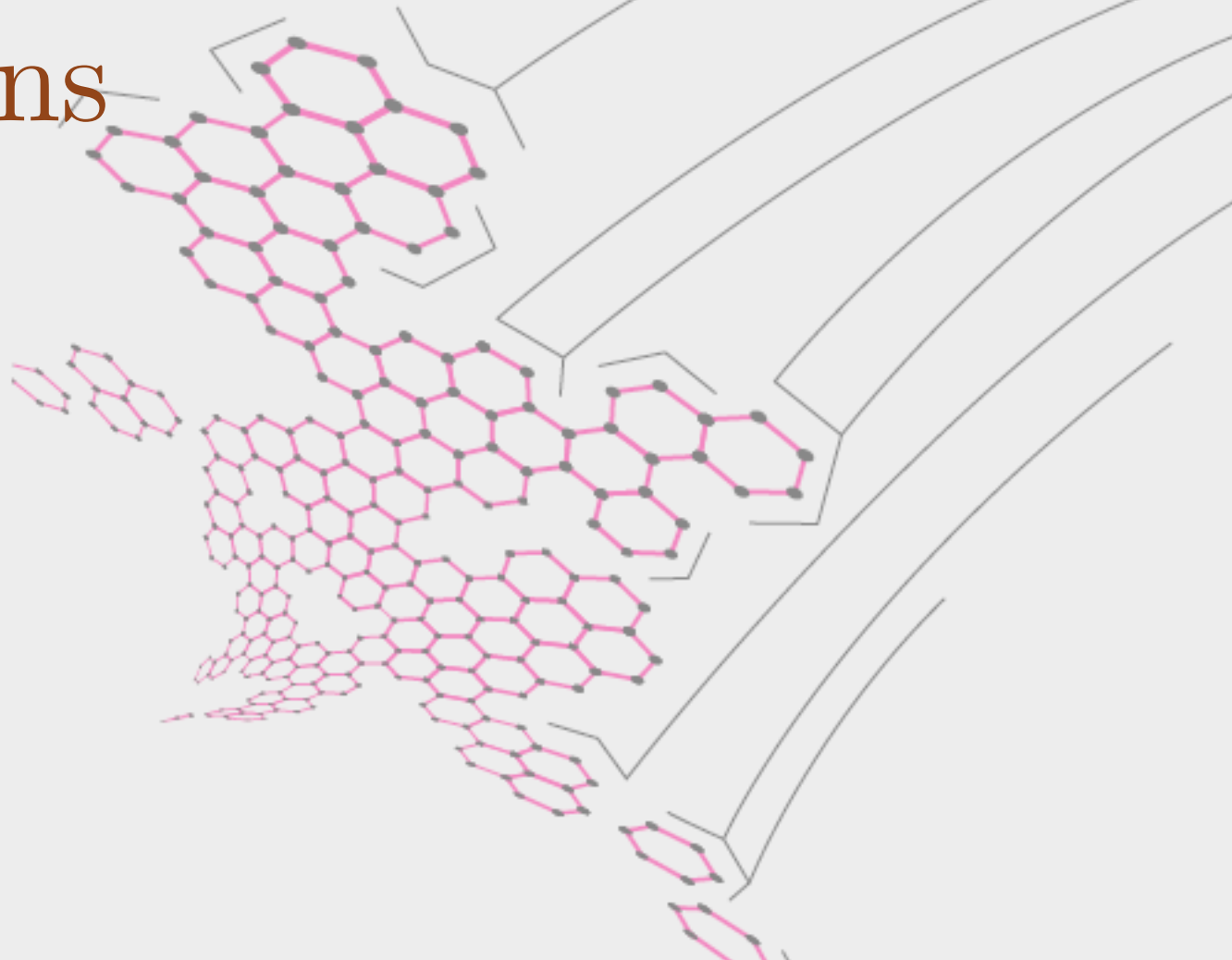
# Calculus A: Lecture 2

## Differentiation of single-variable functions

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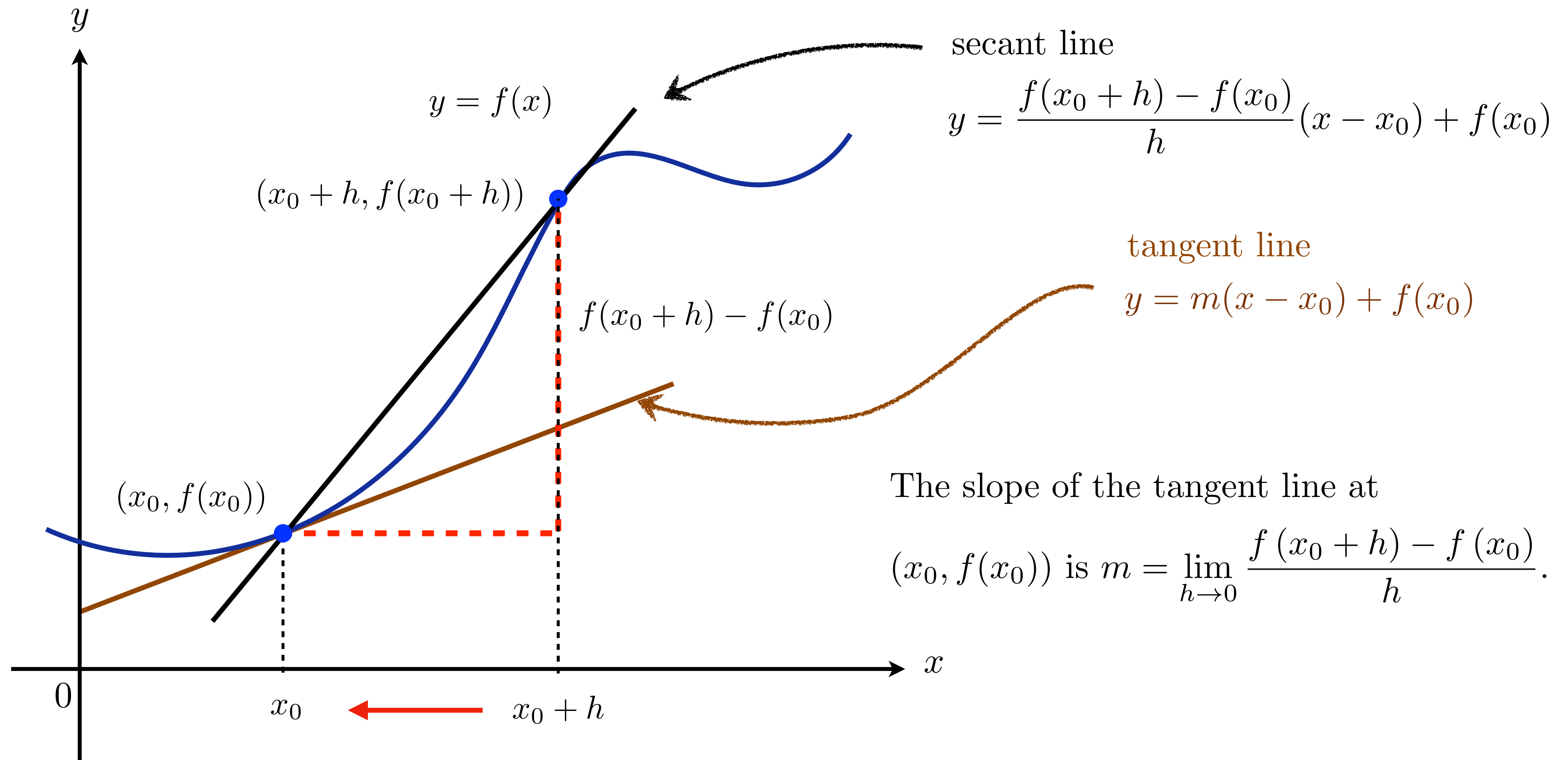
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# Finding a Tangent to the Graph of a Function

To understand how a function changes at a point, we measure the **slope** of its curve there, the **slope of the tangent line**, which leads us to the definition of the derivative.



# Finding a Tangent to the Graph of a Function

**Definition:** The **slope of the curve**  $y = f(x)$  at the point  $P = (x_0, f(x_0))$  is the number

$$m = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} \quad (\text{provided the limit exists}).$$

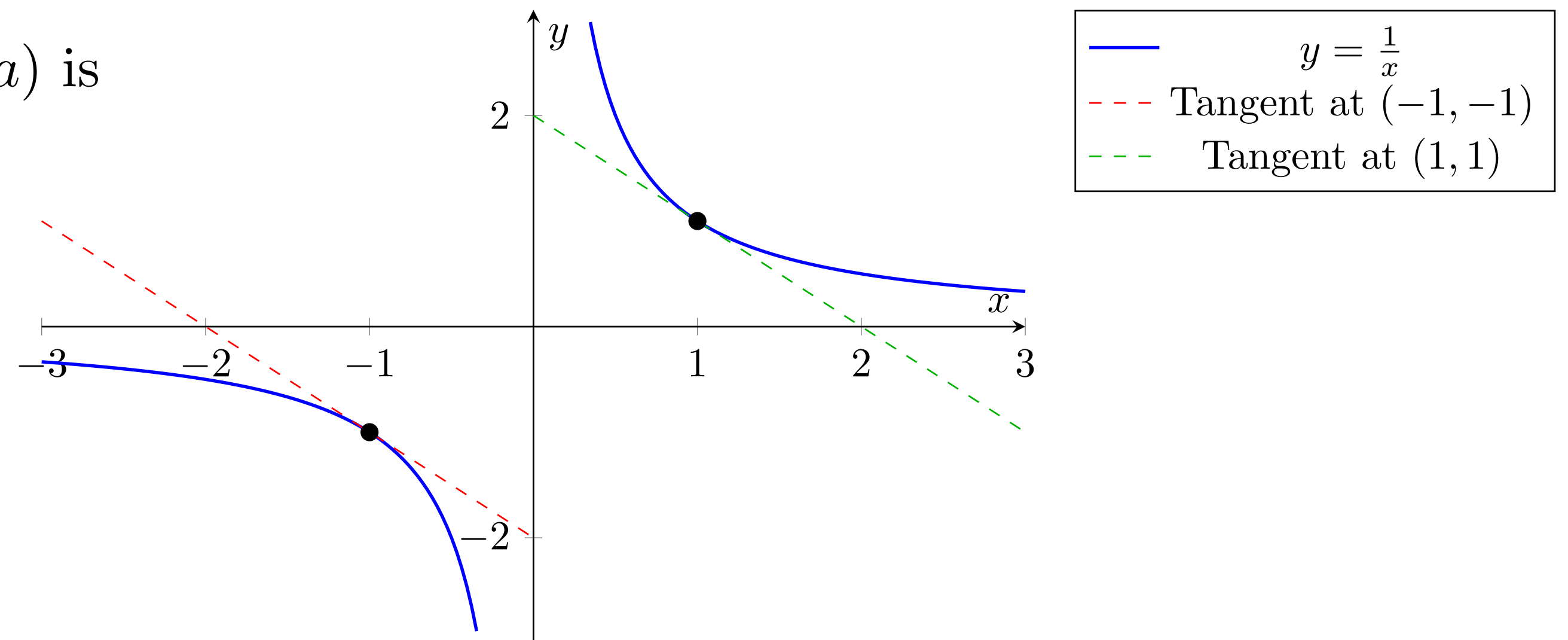
The **tangent line** to the curve at  $P$  is the line through  $P$  with this slope.

**Example:** Find the slope of the curve  $y = 1/x$  at any point  $x = a \neq 0$ .

What is the slope at the point  $x = -1$  ?

**Solution:** For  $f(x) = 1/x$ , the slope at  $(a, 1/a)$  is

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} &= \lim_{h \rightarrow 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h} \\ &= \lim_{h \rightarrow 0} \frac{-1}{a(a+h)} \\ &= -\frac{1}{a^2}. \end{aligned}$$



Here  $a \neq 0$  (positive or negative). At  $a = -1$ , the slope is  $-1$ .

# Derivative at a Point

**Definition:** The **derivative** of a function  $f$  at a point  $x_0$ , denoted  $f'(x_0)$ , is

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

provided this limit exists.

**Definition:** The **derivative** of the function  $f(x)$  with respect to the variable  $x$  is the **function**  $f'$  whose value at  $x$  is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}, \quad \text{(in Leibniz notation: } \frac{dy}{dx} \text{ where } y = f(x).)$$

provided the limit exists.

**Derivative of a Constant Function:** If  $f$  has the constant value  $f(x) = c$ , then

$$\frac{df}{dx} = \frac{d}{dx}(c) = 0$$

**Proof:** If  $f(x) = c$  is constant, then by definition  $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0$ .

# Rules of Differentiation

**Power Rule for Positive Integers:** If  $n$  is a positive integer, then

$$\frac{d}{dx}x^n = nx^{n-1}$$

**Proof** (Power Rule,  $n \in \mathbb{Z}^+$ ):

$$\frac{d}{dx}x^n = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} = \lim_{h \rightarrow 0} \frac{\sum_{k=0}^n \binom{n}{k} x^{n-k} h^k - x^n}{h} = \lim_{h \rightarrow 0} \frac{nx^{n-1}h + \dots}{h} = nx^{n-1}.$$

**Example:** Find the derivative of  $f(x) = \sqrt{x}$  for  $x > 0$ .

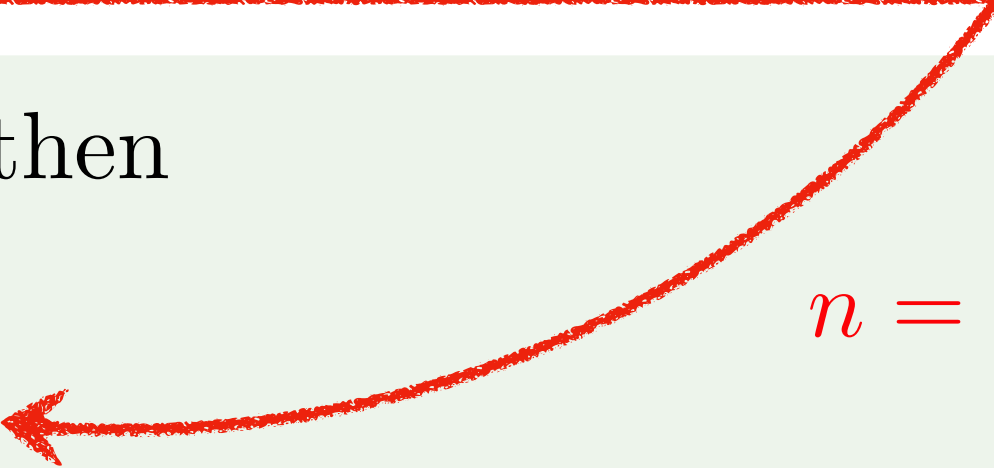
**Solution:**

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} = \lim_{h \rightarrow 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}}.$$

**Power Rule (General Version):** If  $n$  is any real number, then

$$\frac{d}{dx}x^n = nx^{n-1}$$

$n = 1/2$



for all  $x$  where the powers  $x^n$  and  $x^{n-1}$  are defined.

# Rules of Differentiation

**Derivative Sum Rule:** If  $u$  and  $v$  are differentiable functions of  $x$ , then their sum  $u + v$  is differentiable at every point where  $u$  and  $v$  are both differentiable. At such points,

$$\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}$$

**Proof (Sum Rule):**

If  $u$  and  $v$  are differentiable at  $x$ , then

$$\frac{d}{dx}(u+v) = \lim_{h \rightarrow 0} \frac{(u(x+h) + v(x+h)) - (u(x) + v(x))}{h} = \lim_{h \rightarrow 0} \frac{u(x+h) - u(x)}{h} + \frac{v(x+h) - v(x)}{h} = u'(x) + v'(x).$$

**Example:** Find the derivative of  $y = x^3 + \sqrt{x^{2+\pi}}$

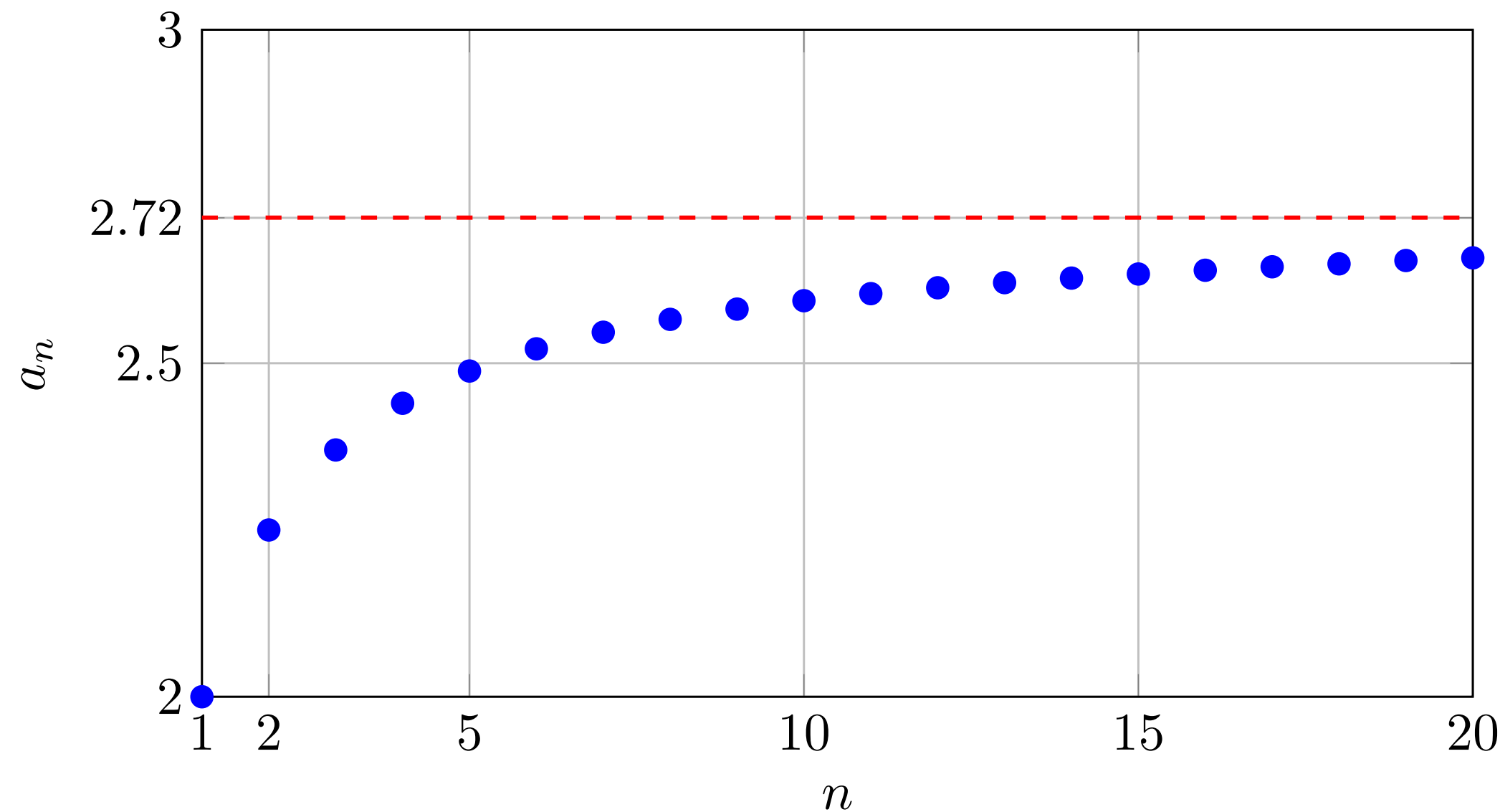
**Solution:**

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(x^3) + \frac{d}{dx}\left(\sqrt{x^{2+\pi}}\right) \\ &= 3x^2 + \frac{d}{dx}\left(x^{\frac{2+\pi}{2}}\right) \\ &= 3x^2 + \frac{2+\pi}{2} x^{\frac{2+\pi}{2}-1} \\ &= 3x^2 + \frac{2+\pi}{2} x^{\frac{\pi}{2}}. \end{aligned}$$

# The number $e$ and the derivative of $e^x$

- Definition of  $e$ :

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n \approx 2.71828$$



$$a_n = \left(1 + \frac{1}{n}\right)^n, \quad n = 1, 2, 3, \dots$$

- Exponential Function:

$$f(x) = e^x, \quad x \in \mathbb{R} = (-\infty, \infty)$$

the derivative of the function  
is the same as the function  
itself

- Derivative (using the limit definition):

$$\frac{d}{dx}e^x = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = \lim_{h \rightarrow 0} \frac{e^x(e^h - 1)}{h} = e^x \underbrace{\lim_{h \rightarrow 0} \frac{e^h - 1}{h}}_{=1} = e^x$$

# Rules of Differentiation

**Derivative Product Rule:** If  $u$  and  $v$  are differentiable at  $x$ , then so is their product  $uv$ , and

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

**Proof (Product Rule):** If  $u$  and  $v$  are differentiable at  $x$ , then

$$\frac{d}{dx}(uv) = \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x)v(x)}{h}.$$

Add and subtract  $u(x+h)v(x)$ :

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x+h)v(x) + u(x+h)v(x) - u(x)v(x)}{h} \\ &= \lim_{h \rightarrow 0} \left[ u(x+h) \frac{v(x+h) - v(x)}{h} + v(x) \frac{u(x+h) - u(x)}{h} \right] = u(x)v'(x) + v(x)u'(x). \end{aligned}$$

**Example:** Find the derivative of  $y = \frac{1}{x} (x^2 + e^x)$ .

**Solution:**  $u = \frac{1}{x}, \quad v = x^2 + e^x$

$$u' = -\frac{1}{x^2}, \quad v' = 2x + e^x$$

$$\Rightarrow \frac{dy}{dx} = u'v + uv' = -\frac{1}{x^2}(x^2 + e^x) + \frac{1}{x}(2x + e^x) = 1 + \frac{e^x(x-1)}{x^2}.$$

# Rules of Differentiation

**Derivative Quotient Rule:** If  $u$  and  $v$  are differentiable at  $x$  and if  $v(x) \neq 0$ , then the quotient  $u/v$  is differentiable at  $x$ , and

$$\frac{d}{dx} \left( \frac{u}{v} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

**Proof (Quotient Rule)**

$$\begin{aligned} \left( \frac{u}{v} \right)' (x) &= \lim_{h \rightarrow 0} \frac{\frac{u(x+h)}{v(x+h)} - \frac{u(x)}{v(x)}}{h} = \lim_{h \rightarrow 0} \frac{u(x+h)v(x) - u(x)v(x+h)}{h v(x+h)v(x)} \\ &= \lim_{h \rightarrow 0} \frac{v(x)(u(x+h) - u(x)) - u(x)(v(x+h) - v(x))}{h v(x+h)v(x)} \\ &= \frac{u'(x)v(x)}{[v(x)]^2} - \frac{u(x)v'(x)}{[v(x)]^2} = \frac{u'(x)v(x) - u(x)v'(x)}{[v(x)]^2}. \end{aligned}$$

**Example:** Find the derivative of  $y = \frac{x^2 - 1}{x^3 + 1}$ .

**Solution:**

$$\begin{aligned} u &= x^2 - 1, & v &= x^3 + 1 \\ u' &= 2x, & v' &= 3x^2 \end{aligned} \quad \implies \quad \frac{dy}{dx} = \frac{u'v - uv'}{v^2} = \frac{-x^4 + 3x^2 + 2x}{(x^3 + 1)^2}.$$

# Trigonometric Functions (Recap)

$$\sin y = \frac{\text{opposite}}{\text{hypotenuse}}$$

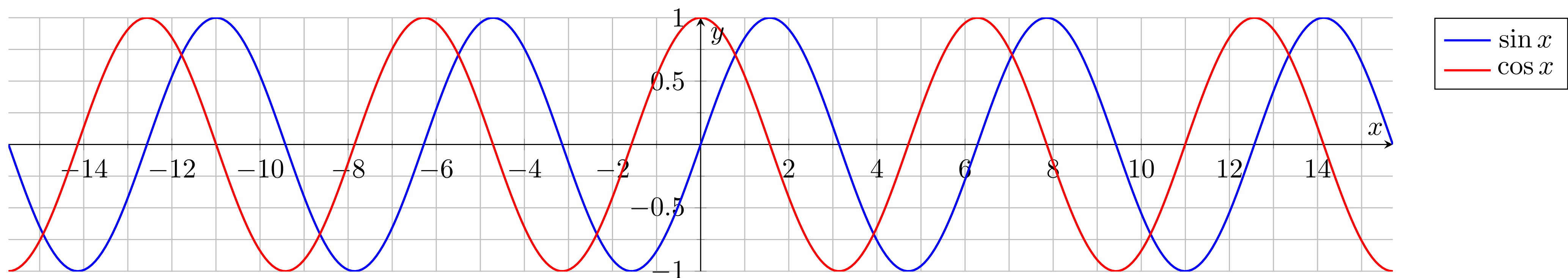
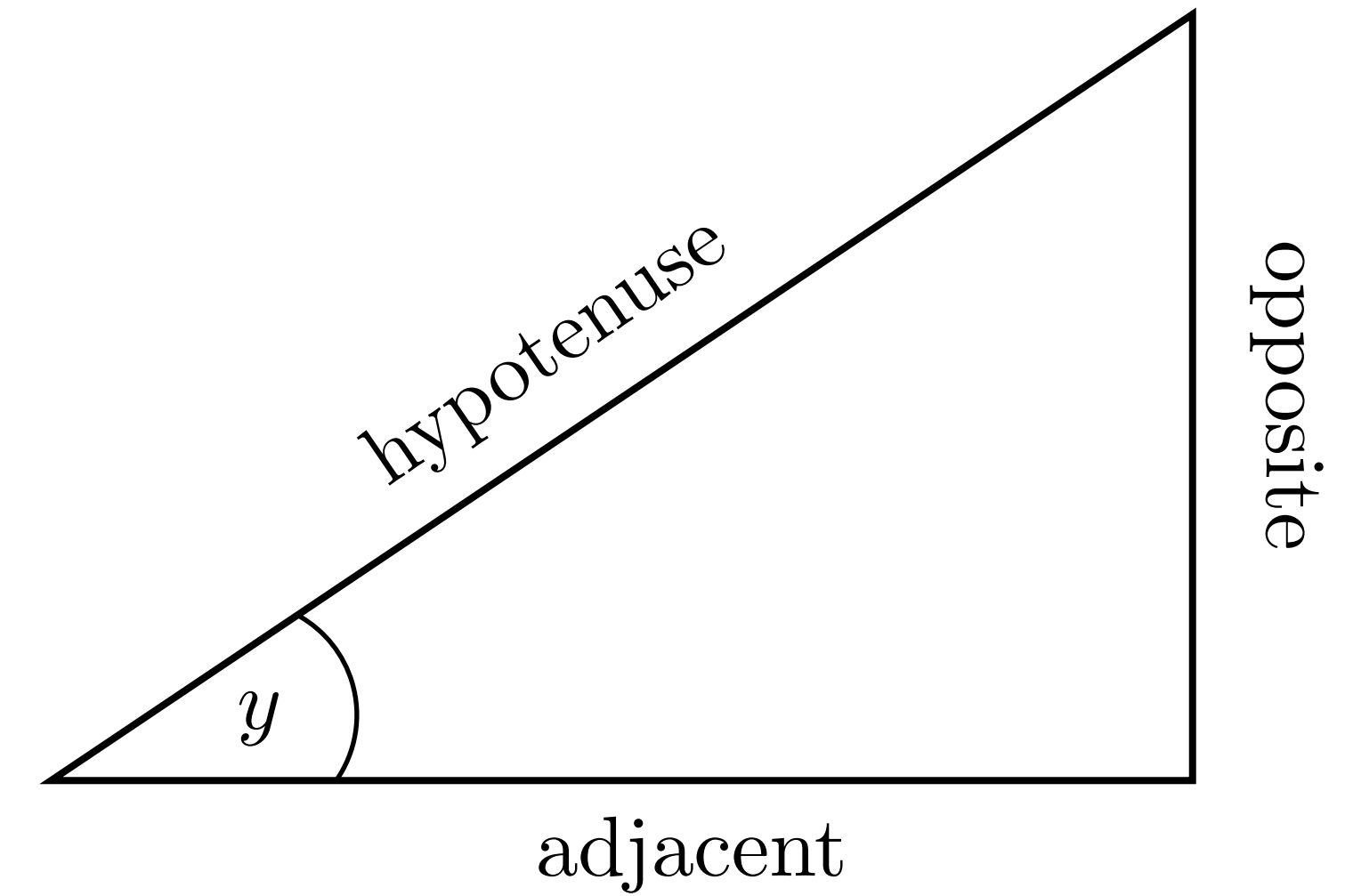
$$\cos y = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan y = \frac{\sin y}{\cos y}$$

$$\cot y = \frac{\cos y}{\sin y}$$

$$\sec y = \frac{1}{\cos y}$$

$$\csc y = \frac{1}{\sin y}$$



# Derivatives of Trigonometric Functions

The derivative of the sine function is the cosine function:  $\frac{d}{dx}(\sin x) = \cos x$

The derivative of the cosine function is the negative of the sine function:  $\frac{d}{dx}(\cos x) = -\sin x$

The derivatives of the other trigonometric functions:

$$\begin{aligned}\frac{d}{dx}(\tan x) &= \sec^2 x & \frac{d}{dx}(\cot x) &= -\csc^2 x \\ \frac{d}{dx}(\sec x) &= \sec x \tan x & \frac{d}{dx}(\csc x) &= -\csc x \cot x\end{aligned}$$

where:

$$\begin{aligned}\sec x &= \frac{1}{\cos x}, & x &\neq \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}, & \csc x &= \frac{1}{\sin x}, & x &\neq k\pi, \quad k \in \mathbb{Z}, \\ \tan x &= \frac{\sin x}{\cos x}, & x &\neq \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z}, & \cot x &= \frac{\cos x}{\sin x}, & x &\neq k\pi, \quad k \in \mathbb{Z}.\end{aligned}$$

# The Chain Rule

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How do we differentiate  $F(x) = e^{3x^2}$ ? This function is a composition:  $y = f(u) = e^u$  and  $u = g(x) = 3x^2$ , both easy to differentiate. The Chain Rule tells us how.

**Theorem** (The Chain Rule): If  $f(u)$  is differentiable at the point  $u = g(x)$  and  $g(x)$  is differentiable at  $x$ , then the composite function  $(f \circ g)(x) = f(g(x))$  is differentiable at  $x$ , and

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

In Leibniz's notation, if  $y = f(u)$  and  $u = g(x)$ , then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx},$$

where  $dy/du$  is evaluated at  $u = g(x)$ .

**Example:** Differentiate  $\sin(x^2 + e^x)$  with respect to  $x$ .

**Solution:** Using the chain rule, let  $f(u) = \sin(u)$  and  $g(x) = x^2 + e^x$ . Then

$$\frac{dy}{dx} = \underbrace{\cos(g(x))}_{f'(g(x))} \frac{dg}{dx} = \cos(x^2 + e^x) \cdot (2x + e^x).$$

# The Chain Rule

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**Proof (Chain Rule):** Suppose  $y = f(u)$  and  $u = g(x)$  are differentiable. Then

$$\frac{d}{dx} f(g(x)) = \lim_{h \rightarrow 0} \frac{f(g(x+h)) - f(g(x))}{h}.$$

Set  $\Delta u = g(x+h) - g(x)$ , so that  $\lim_{h \rightarrow 0} \Delta u \rightarrow 0$ . Then

$$\frac{f(g(x+h)) - f(g(x))}{h} = \frac{f(g(x) + \Delta u) - f(g(x))}{\Delta u} \cdot \frac{\Delta u}{h}.$$

Taking the limit as  $h \rightarrow 0$ :

$$\lim_{h \rightarrow 0} \frac{f(g(x+h)) - f(g(x))}{h} = \underbrace{\lim_{\Delta u \rightarrow 0} \frac{f(g(x) + \Delta u) - f(g(x))}{\Delta u}}_{f'(g(x))} \cdot \underbrace{\lim_{h \rightarrow 0} \frac{\Delta u}{h}}_{g'(x)} = f'(g(x)) g'(x).$$

# Implicit Differentiation

# Implicit Differentiation

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Sometimes  $y$  is not given explicitly as a function of  $x$ , but only implicitly by an equation like  $F(x, y) = 0$ ; implicit differentiation lets us find  $dy/dx$  even in these cases.

**Example:** The equation

$$x^2 + y^2 = 25$$

implicitly defines a function  $y(x)$ . Find  $\frac{dy}{dx}$ .

**Solution:** Differentiate both sides with respect to  $x$ :

$$2x + 2y \frac{dy}{dx} = 0 \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{x}{y}.$$

This illustrates how implicit differentiation allows us to compute derivatives even when  $y$  is not explicitly solved as a function of  $x$ .

**Implicit differentiation procedure:**

1. Differentiate both sides of the equation with respect to  $x$ , treating  $y$  as a function of  $x$ .
2. Collect all terms involving  $\frac{dy}{dx}$  on one side and solve for  $\frac{dy}{dx}$ .

# Derivatives of Inverse Functions

**Definition** (Inverse Function) A function  $f$  has an **inverse**  $f^{-1}$  if for every  $y$  in the range of  $f$ , there exists a unique  $x$  such that

$$f(x) = y \quad \text{and} \quad f^{-1}(y) = x.$$

**Key property:**  $f(f^{-1}(y)) = y$  and  $f^{-1}(f(x)) = x$

**Theorem** (The Derivative Rule for Inverses) If  $f$  has an interval  $I$  as domain and  $f'(x)$  exists and is never zero on  $I$ , then  $f^{-1}$  is differentiable at every point in its domain (the range of  $f$ ). The value of  $(f^{-1})'$  at a point  $b$  in the domain of  $f^{-1}$  is the reciprocal of the value of  $f'$  at the point  $a = f^{-1}(b)$  :

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}$$

or

$$\left. \frac{df^{-1}}{dx} \right|_{x=b} = \frac{1}{\left. \frac{df}{dx} \right|_{x=f^{-1}(b)}}$$

# Derivatives of Inverse Functions

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**Main idea behind the theorem:**

$$f(f^{-1}(x)) = x$$

Inverse function relationship

$$\frac{d}{dx} f(f^{-1}(x)) = 1$$

Differentiate both sides

$$f'(f^{-1}(x)) \cdot \frac{d}{dx} f^{-1}(x) = 1$$

Chain Rule

$$\frac{d}{dx} f^{-1}(x) = \frac{1}{f'(f^{-1}(x))}$$

Solve for the derivative

**Example:** The function  $f(x) = x^2$ ,  $x \geq 0$ , has inverse  $f^{-1}(x) = \sqrt{x}$  with derivatives

$$f'(x) = 2x, \quad (f^{-1})'(x) = \frac{1}{2\sqrt{x}}.$$

Using the theorem:

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} = \frac{1}{2(f^{-1}(x))} = \frac{1}{2\sqrt{x}},$$

which agrees with the known derivative.

# Derivative of the Natural Logarithm Function

The **natural logarithm**  $\ln(x)$  is the **inverse** of the exponential function  $e^x$ :

$$y = \ln(x) \quad \text{means} \quad e^y = x, \quad x > 0$$

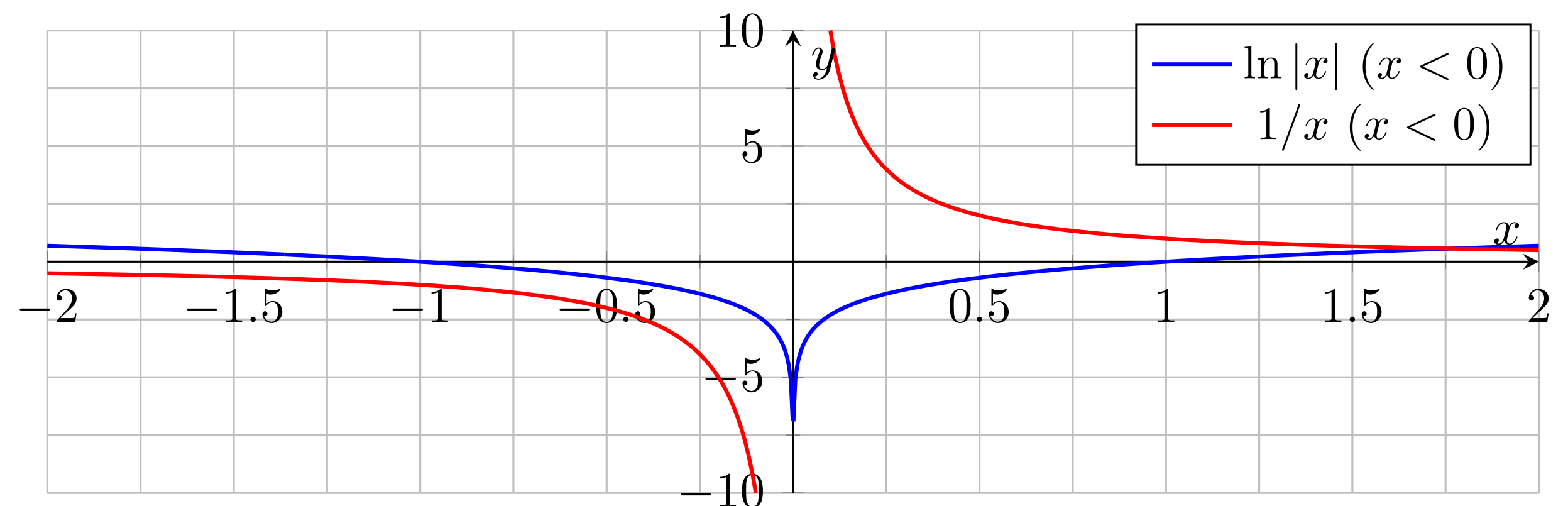
**Key properties:**

$$\ln(xy) = \ln(x) + \ln(y), \quad \ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y), \quad \ln(x^r) = r \ln(x)$$

Since the exponential function  $f(x) = e^x$  is differentiable everywhere, we can use the theorem to find the derivative of its inverse  $f^{-1}(x) = \ln x$ :

$$(f^{-1})'(x) = \underbrace{\frac{1}{f'(f^{-1}(x))}}_{\text{use theorem}} = \underbrace{\frac{1}{e^{f^{-1}(x)}}}_{\text{use } f'(u)=e^u} = \frac{1}{e^{\ln x}} = \frac{1}{x} \quad \text{using inverse function relationship}$$

It can be shown that  $\frac{d}{dx} \ln|x| = \frac{1}{x}, \quad x \neq 0$



# Inverse Trigonometric Functions

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**Tangent function:** For  $y \in \mathbb{R}$  with  $\cos y \neq 0$ ,

$$\tan y = \frac{\sin y}{\cos y}.$$

**Derivative of  $y = \tan^{-1} x$  using implicit differentiation:**

inverse tangent  
function



$$y = \tan^{-1} x \quad (\tan y = x)$$

$$\frac{d}{dx}(\tan y) = \frac{d}{dx}(x) \quad (\text{differentiate both sides})$$

$$\sec^2 y \cdot \frac{dy}{dx} = 1 \quad (\text{chain Rule})$$

$$\frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$= \frac{1}{1 + \tan^2 y} \quad (\text{use } \sec^2 y = 1 + \tan^2 y)$$

$$= \frac{1}{1 + x^2} \quad (\text{since } \tan y = x)$$

**Conclusion:**

$$\frac{d}{dx} \tan^{-1} x = \frac{1}{1 + x^2}$$

# Trigonometric and Inverse Trigonometric Functions

| Function      | Definition                        | Domain                           | Range                       | Derivative                   |
|---------------|-----------------------------------|----------------------------------|-----------------------------|------------------------------|
| $\sin^{-1} x$ | $y = \sin^{-1} x \iff \sin y = x$ | $[-1, 1]$                        | $[-\pi/2, \pi/2]$           | $\frac{1}{\sqrt{1-x^2}}$     |
| $\cos^{-1} x$ | $y = \cos^{-1} x \iff \cos y = x$ | $[-1, 1]$                        | $[0, \pi]$                  | $-\frac{1}{\sqrt{1-x^2}}$    |
| $\tan^{-1} x$ | $y = \tan^{-1} x \iff \tan y = x$ | $(-\infty, \infty)$              | $(-\pi/2, \pi/2)$           | $\frac{1}{1+x^2}$            |
| $\cot^{-1} x$ | $y = \cot^{-1} x \iff \cot y = x$ | $(-\infty, \infty)$              | $(0, \pi)$                  | $-\frac{1}{1+x^2}$           |
| $\sec^{-1} x$ | $y = \sec^{-1} x \iff \sec y = x$ | $(-\infty, -1] \cup [1, \infty)$ | $[0, \pi], y \neq \pi/2$    | $\frac{1}{ x \sqrt{x^2-1}}$  |
| $\csc^{-1} x$ | $y = \csc^{-1} x \iff \csc y = x$ | $(-\infty, -1] \cup [1, \infty)$ | $[-\pi/2, \pi/2], y \neq 0$ | $-\frac{1}{ x \sqrt{x^2-1}}$ |

where:

$$\sec x = \frac{1}{\cos x}, \quad x \neq \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z},$$

$$\tan x = \frac{\sin x}{\cos x}, \quad x \neq \frac{\pi}{2} + k\pi, \quad k \in \mathbb{Z},$$

$$\csc x = \frac{1}{\sin x}, \quad x \neq k\pi, \quad k \in \mathbb{Z},$$

$$\cot x = \frac{\cos x}{\sin x}, \quad x \neq k\pi, \quad k \in \mathbb{Z}.$$

# Extreme Values of Functions

# Extreme Values of Functions

**Definition** Let  $f$  be a function with domain  $D$ . Then  $f$  has an **absolute maximum** value on  $D$  at a point  $c$  if

$$f(x) \leq f(c) \quad \text{for all } x \text{ in } D$$

and an **absolute minimum** value on  $D$  at  $c$  if

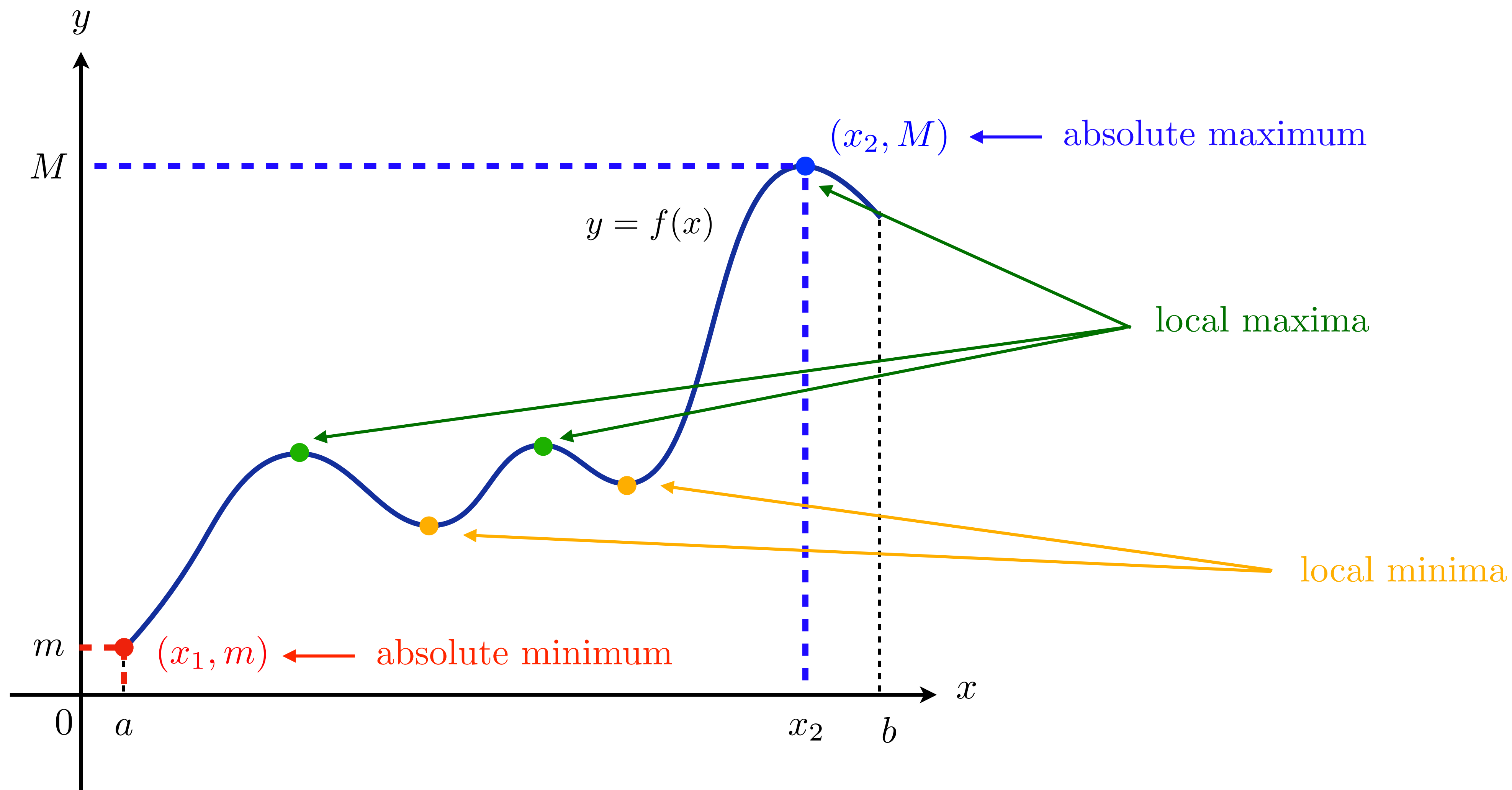
$$f(x) \geq f(c) \quad \text{for all } x \text{ in } D.$$

**Theorem** (The Extreme Value Theorem): If  $f$  is **continuous** on a closed interval  $[a, b]$ , then  $f$  attains both an **absolute maximum** value  $M$  and an **absolute minimum** value  $m$  in  $[a, b]$ . That is, there are numbers  $x_1$  and  $x_2$  in  $[a, b]$  with  $f(x_1) = m$ ,  $f(x_2) = M$ , and  $m \leq f(x) \leq M$  for every other  $x$  in  $[a, b]$ .

**Definition:** A function  $f$  has a **local maximum** value at a point  $c$  within its domain  $D$  if  $f(x) \leq f(c)$  for all  $x \in D$  lying in some open interval containing  $c$ .

A function  $f$  has a **local minimum** value at a point  $c$  within its domain  $D$  if  $f(x) \geq f(c)$  for all  $x \in D$  lying in some open interval containing  $c$ .

# Extreme Values of Functions



# Extreme Values of Functions

**Theorem** (The First Derivative Theorem for Local Extreme Values): If  $f$  has a **local maximum** or **local minimum** value at an interior point  $c$  of its domain, and if  $f'$  is defined at  $c$ , then

$$f'(c) = 0$$

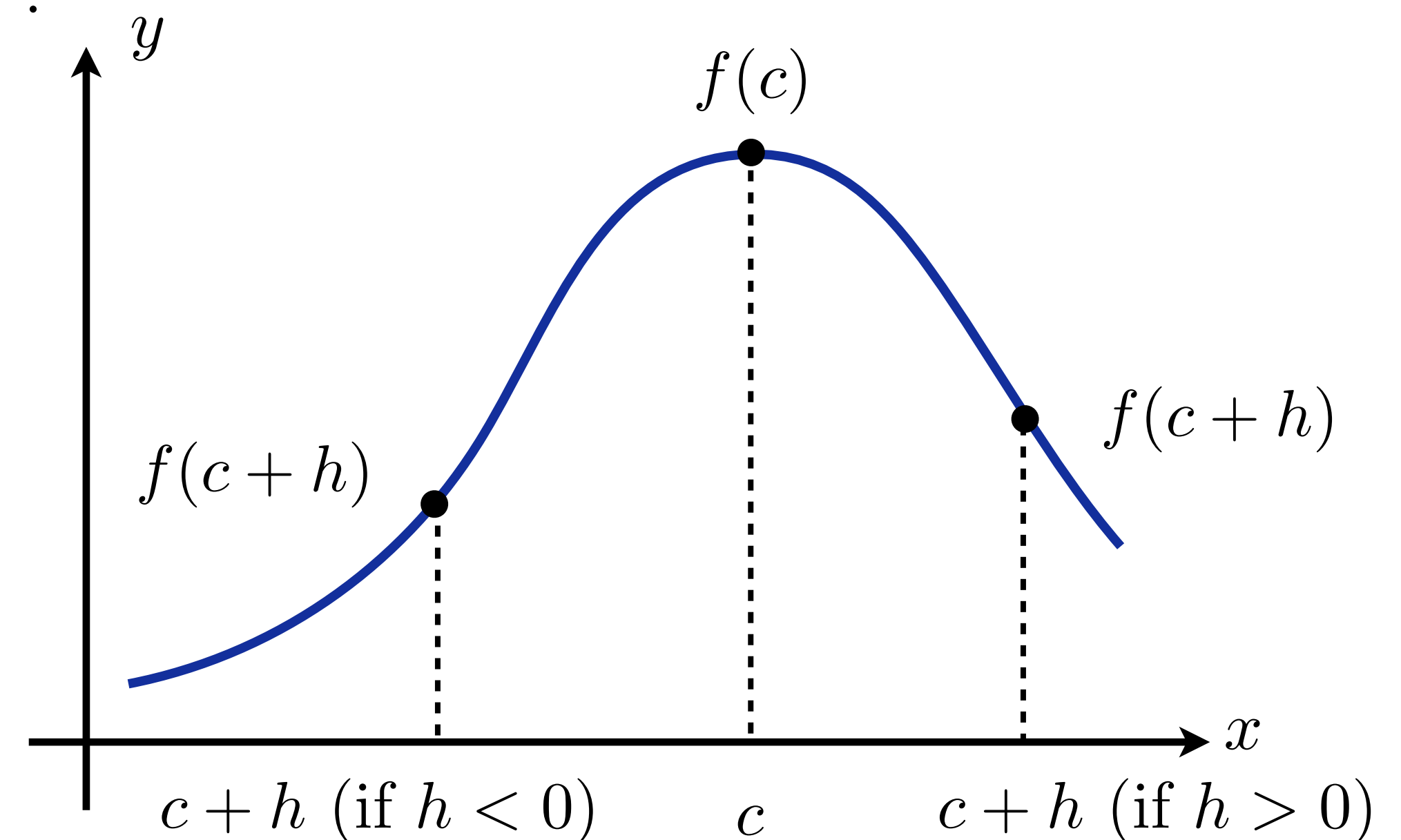
**Proof (sketch).** Assume  $f$  has a **local maximum** at  $c$ , so for  $x$  near  $c$  we have  $f(x) \leq f(c)$ .  
The derivative is

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}.$$

- For  $h > 0$ :  $f(c+h) - f(c) \leq 0 \implies \frac{f(c+h) - f(c)}{h} \leq 0$ .
- For  $h < 0$ :  $f(c+h) - f(c) \leq 0 \implies \frac{f(c+h) - f(c)}{h} \geq 0$ .

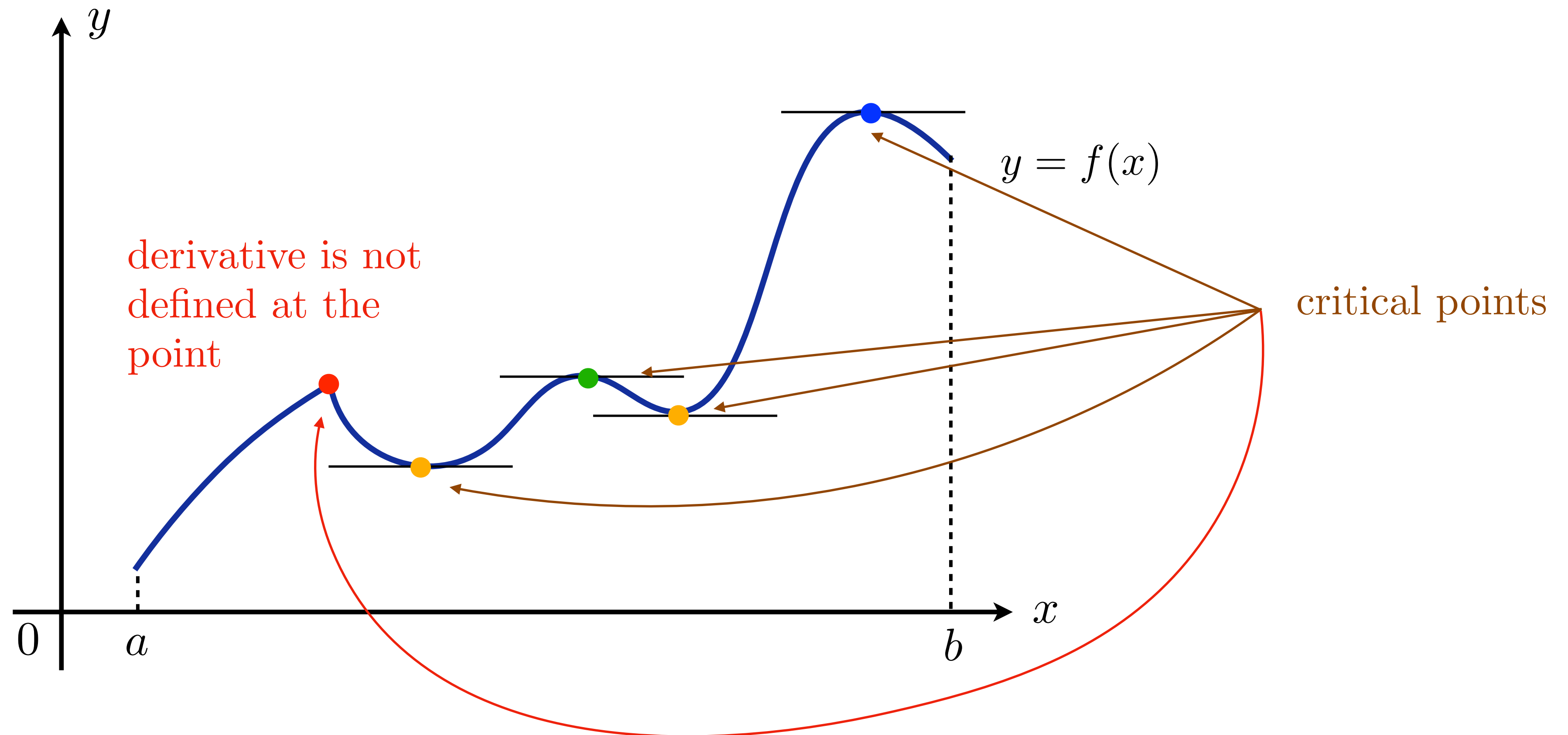
Taking the limit from both sides gives a derivative  $\geq 0$  and  $\leq 0$ .

Since  $f'(c)$  exists,  $f'(c) = 0$ .



# Extreme Values of Functions

**Definition:** An interior point of the domain of a function  $f$  where  $f'$  is zero or **undefined** is a **critical point** of  $f$ .



# Extreme Values of Functions

## Finding Absolute Extrema on a Closed Interval:

1. Evaluate  $f$  at all critical points and at the endpoints of the interval.
2. The largest value gives the absolute maximum; the smallest gives the absolute minimum.

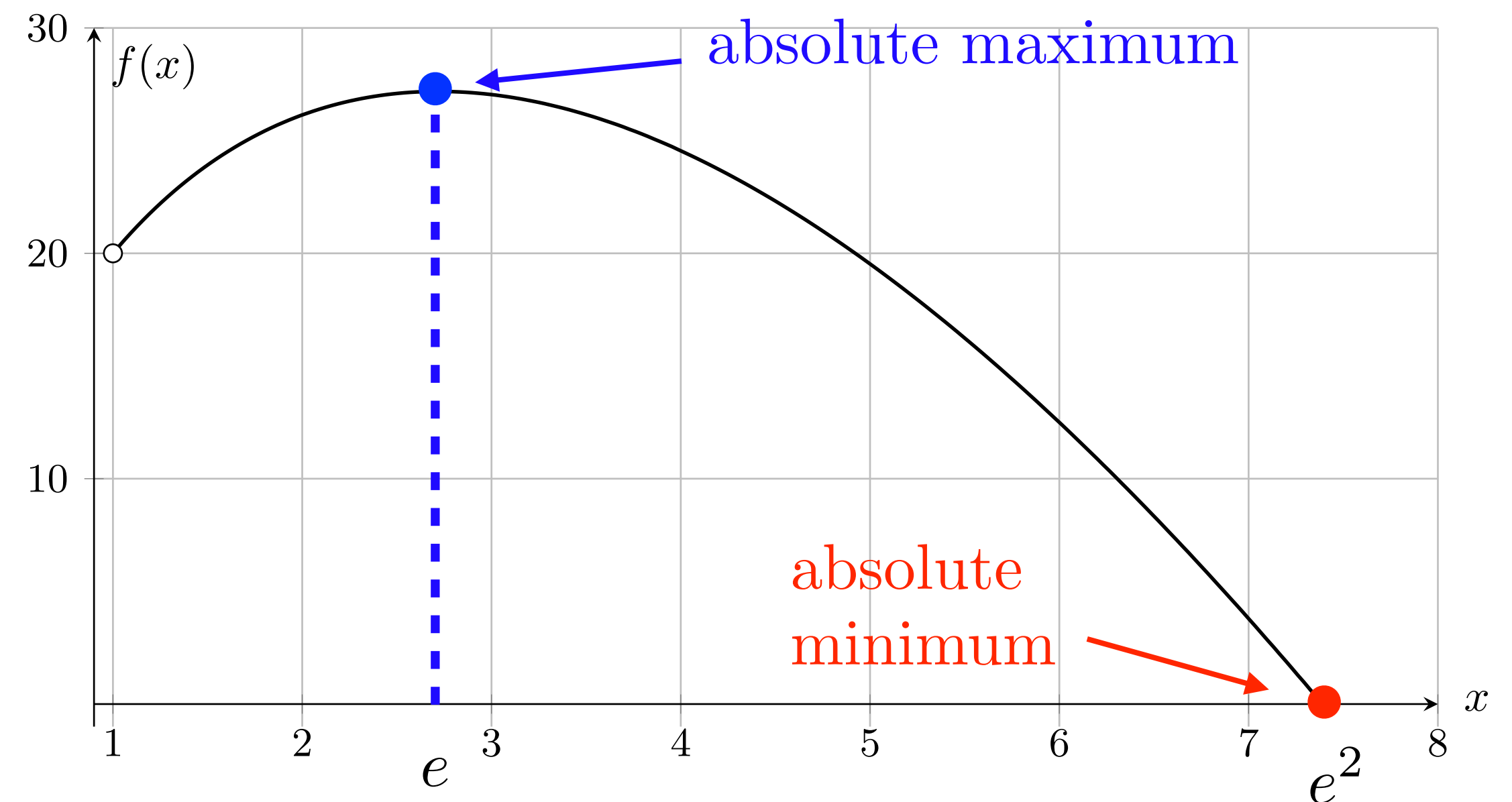
**Example:** Find the absolute maximum and minimum values of  $f(x) = 10x(2 - \ln x)$  on the interval  $[1, e^2]$ .

**Solution.**

$$f'(x) = 10(1 - \ln x) = 0 \implies x = e \text{ (critical point)}$$

$$f(1) = 20, \quad f(e) = 10e, \quad f(e^2) = 0.$$

Hence the absolute maximum is  $f(e) = 10e$  at  $x = e$ , and the absolute minimum is  $f(e^2) = 0$  at  $x = e^2$ .



# Extreme Values of Functions

**Monotonicity via Derivative:** Let  $f$  be continuous on  $[a, b]$  and differentiable on  $(a, b)$ .

- $f$  is **increasing** on  $[a, b]$  if for all  $x_1, x_2 \in [a, b]$  with  $x_1 < x_2$ , we have

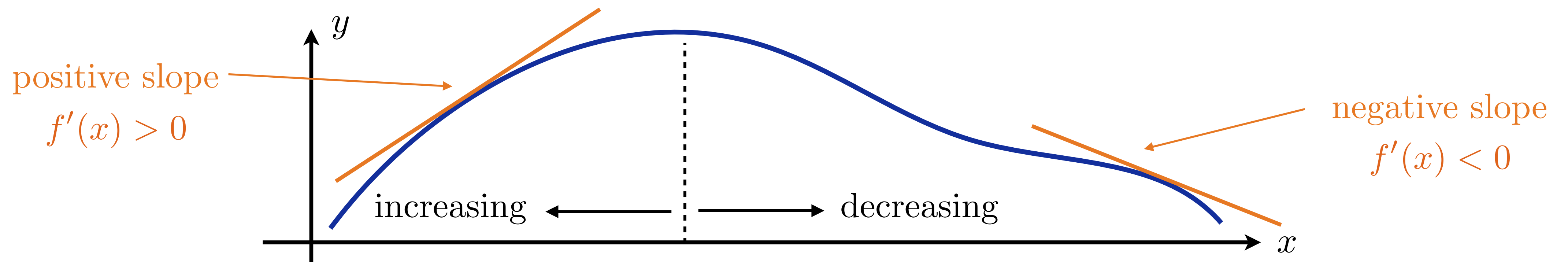
$$f(x_1) < f(x_2).$$

If  $f'(x) > 0$  for all  $x \in (a, b)$ , then  $f$  is increasing on  $[a, b]$ .

- $f$  is **decreasing** on  $[a, b]$  if for all  $x_1, x_2 \in [a, b]$  with  $x_1 < x_2$ , we have

$$f(x_1) > f(x_2).$$

If  $f'(x) < 0$  for all  $x \in (a, b)$ , then  $f$  is decreasing on  $[a, b]$ .



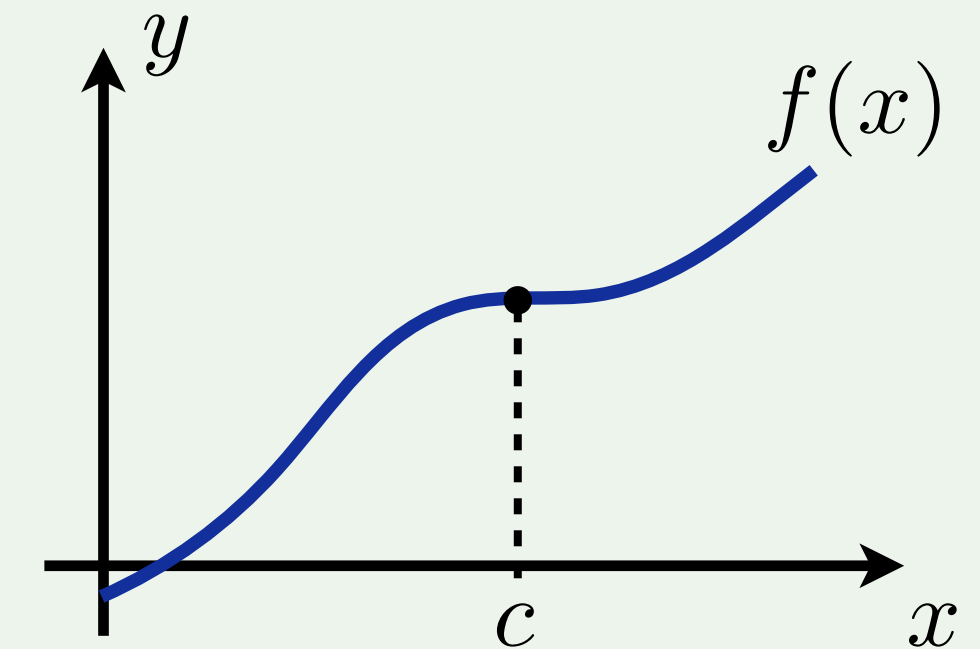
A function that is increasing or decreasing on an interval is called **monotonic**.

# Extreme Values of Functions

## First Derivative Test for Local Extrema:

Let  $c$  be a **critical point** of a continuous function  $f$ , differentiable on an interval around  $c$  (except possibly at  $c$ ).

1. If  $f'$  changes from negative to positive at  $c$ ,  $f$  **has a local minimum at  $c$** .
2. If  $f'$  changes from positive to negative at  $c$ ,  $f$  **has a local maximum at  $c$** .
3. If  $f'$  does not change sign at  $c$ ,  $f$  **has no local extremum at  $c$** .



**Example:** Find the critical points of  $f(x) = x^{1/3}(x - 4) = x^{4/3} - 4x^{1/3}$ . Identify the intervals on which  $f$  is increasing and decreasing. Find the function's local and absolute extreme values.

**Solution.**

$$f'(x) = \frac{4(x - 1)}{3x^{2/3}}$$

vanishes at  $x = 1$ ; undefined at  $x = 0$ .

With no endpoints.

Extreme values occur at  $x = 0$  and  $x = 1$ .

| Interval        | $x < 0$    | $0 < x < 1$ | $x > 1$    |
|-----------------|------------|-------------|------------|
| Sign of $f'$    | —          | —           | +          |
| Behavior of $f$ | decreasing | decreasing  | increasing |

# L'Hôpital's Rule

# L'Hôpital's Rule

**Theorem** (L'Hôpital's Rule): Suppose that  $f(a) = g(a) = 0$ , that  $f$  and  $g$  are differentiable on an open interval  $I$  containing  $a$ , and that  $g'(x) \neq 0$  on  $I$  if  $x \neq a$ . Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

assuming that the limit on the right side of this equation exists.

**Using L'Hôpital's Rule:**

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \quad \text{if the limit is of the form } 0/0 \text{ or } \infty/\infty.$$

Differentiate repeatedly until the limit is defined.



**Do not apply if numerator or denominator has a finite nonzero limit.**

# L'Hôpital's Rule

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**(L'Hôpital's rule idea):** Suppose a limit is of the form “0/0”. Rewrite the fraction as a product:

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{x - a} \cdot \frac{x - a}{g(x) - g(a)}.$$

Since  $f(a) = g(a) = 0$ , this becomes

$$\frac{f(x)}{g(x)} = \frac{f(x) - 0}{x - a} \cdot \frac{x - a}{g(x) - 0} = \frac{f(x) - f(a)}{x - a} \cdot \frac{x - a}{g(x) - g(a)}.$$

Take the limit as  $x \rightarrow a$ :

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} \frac{x - a}{g(x) - g(a)}.$$

By the definition of derivative:

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = f'(a), \quad \lim_{x \rightarrow a} \frac{x - a}{g(x) - g(a)} = \frac{1}{g'(a)}.$$

Multiply them:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)} \quad \text{if } g'(a) \neq 0.$$

# L'Hôpital's Rule

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**Example:** Evaluate the limits:

$$(a) \lim_{x \rightarrow 0} \frac{3x - \sin x}{x}, \quad (b) \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}, \quad (c) \lim_{x \rightarrow \infty} \frac{\ln x}{2\sqrt{x}}$$

**Solution.**

(a)  $\lim_{x \rightarrow 0} \frac{3x - \sin x}{x}$ . Since  $\frac{0}{0}$  form, apply L'Hôpital:

$$\lim_{x \rightarrow 0} \frac{3 - \cos x}{1} = 3 - 1 = 2.$$

(b)  $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$ . Also  $\frac{0}{0}$  form. Apply L'Hôpital three times:

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} \xrightarrow{\text{L'H}} \lim_{x \rightarrow 0} \frac{\sin x}{6x} \xrightarrow{\text{L'H}} \lim_{x \rightarrow 0} \frac{\cos x}{6} = \frac{1}{6}.$$

(c)  $\lim_{x \rightarrow \infty} \frac{\ln x}{2\sqrt{x}}$ .  $\frac{\infty}{\infty}$  form. Apply L'Hôpital:

$$\lim_{x \rightarrow \infty} \frac{1/x}{1/\sqrt{x}} = \lim_{x \rightarrow \infty} \frac{\sqrt{x}}{x} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} = 0.$$

# Antiderivatives

# Antiderivatives

**Definition** A function  $F$  is an **antiderivative** of  $f$  on an interval  $I$  if

$$F'(x) = f(x)$$

for all  $x$  in  $I$ .

**Observation:** If  $F$  is an antiderivative of  $f$  on an interval  $I$ , then the most general antiderivative of  $f$  on  $I$  is

$$F(x) + C$$

where  $C$  is an arbitrary constant.

**Example:** Find an antiderivative of  $f(x) = 3x^2$  that satisfies  $F(1) = -1$ .

**Solution.** An antiderivative of  $f(x) = 3x^2$  is a function  $F(x)$  such that  $F'(x) = 3x^2$ . We know that the derivative of  $x^3$  is  $3x^2$ , so

$$F(x) = x^3 + C.$$

Using  $F(1) = -1$  gives  $1^3 + C = -1 \implies C = -2$ . Therefore, the required antiderivative is

$$F(x) = x^3 - 2.$$